This evidence questions the absolute stability of the southern region. The present seismic disturbances are probably due to recent rock movements derived from Quaternary tectonic activity taking place since the Tertiary period, and seem to be more than could be attributed to improvements in the capability and distribution of the monitoring network in the region. Fig. 1 shows the seismicity map of south India on the basis of information from the Gauribidanur array since 1966. These findings are of sufficient seismological importance to warrant an intensive investigation into the structure of the Deccan Plateau by seismic, geological and geophysical methods. The Gauribidanur array is ideally suited for the deep seismic sounding of this region.

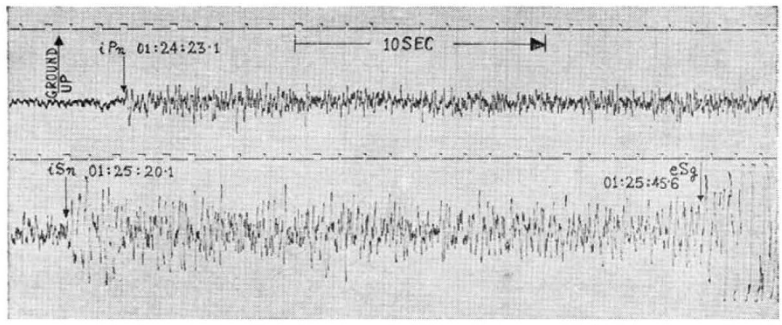


Fig. 3. Reproductions showing P_n, S_n and S_g phases of the same seismogram (Bhadrachalam aftershock of April 15, onset at GBA 0124·23·1) traced by vertical Willmore Mk. II seismometer (natural period: 1 s), 8·75 km away from the array centre. Overall system gain is 183,000 at 1 Hz and chart speed 10 mm s^{-1}. The first and third traces from top represent the time code.

Analysis of the Bhadrachalam earthquake series of April 1969 has thrown some light on the delineation of the Moho beneath the crust in south India, and revealed some interesting features from the velocity of Moho-refracted waves from Bhadrachalam to Gauribidanur. The rate of occurrence of the Bhadrachalam events seems to fall off rapidly, particularly after April 20, although the energy distribution does not seem to follow any regular trend (Fig. 2). About sixty-five aftershocks were recorded at GBA (Gauribidanur array station) over a period of 15 days following the main shock of April 13, followed by one each on May 1 and 2. Thirteen were analysed in detail, and the results are presented in Table 1. A typical aftershock seismogram obtained near the centre point of GBA is shown in Fig. 3.

Table 1. THE BHADRACHALAM EARTHQUAKE SERIES

Shock No.	Date (April 1969)	Onset time at B1* (P_n phase) h	m	s	(a) Apparent vel. (km s^{-1}) (b) Epicentral dist. (deg.) (c) Backbearing† (a)	(b)	(c)	Magnitude
M	13	15	26	11·2	8·96	4·69	36·1	6·1 (S)
1	13	17	04	19·6	8·56	4·70	38·9	3·9
2	13	17	42	50·7	8·48	4·74	44·1	4·2
3	13	17	48	44·8	8·90	4·69	36·0	4·1
4	13	17	55	52·5	8·87	4·79	35·0	4·4
5	15	01	24	23·1	8·77	4·75	38·2	4·4
6	15	17	59	55·4	9·20	4·76	38·3	6·0
7	16	07	42	21·5	8·90	4·80	37·5	4·0
8	16	12	52	48·8	8·60	4·79	37·9	4·3
9	16	19	16	45·6	8·60	4·76	36·0	4·0
10	17	00	06	30·7	8·51	4·75	39·9	4·6
11	19	21	02	54·8	9·20	4·77	38·3	4·0
12	21	09	44	13·1	8·50	4·73	39·9	4·1
13	27	23	14	34·5	9·16	4·73	40·4	4·7

M indicates the main event.
* Cross-over point of the array; location: 13° 36′ 15″ N, 77° 26′ 10″ E.
† Station-to-epicentre azimuth measured clockwise from north.
S shows that the record was saturated even on the attenuated channels operating at an approximate magnification of 22,000 at 1 Hz.

Like previous earthquakes in the Indian plateau, the Bhadrachalam series produced signals characteristic of crustal focus. The depths of these events were less than 20 km. Surface-reflected phases observed at EKA (Eskdalemuir array) and UPP (Uppsala, Sweden) imply a focal depth near 15 km for the main event. Strong Rayleigh-wave coda lasting more than an hour on the long-period records at Gauribidanur also favour a shallow focus. Furthermore, P times at nine observing stations including GBA give a best fit at a mean source depth of 15 km for the main event and one major aftershock (April 15, 1969, onset time 1759·55·4) by the precise epicentre determination method[3] using Herrin's new travel times[4]. The main shock was found to be about 37 km from Bhadrachalam town (geographic location of the town: 17° 40′ 12″ N, 80° 54′ 36″ E) in the direction 25° north of west.

Table 2. ANALYSIS OF FOUR EVENTS

Shock No.	Date	Onset at GBA h	m	s	Region of epicentre	α_n (km s^{-1})
1	March 18, 1968	14	57	40·4	Mahbubabad, Andhra Pradesh	8·91
2	April 25, 1968	20	10	04·3	North-east of Nellore, Andhra Pradesh	8·90
3	May 12, 1968	18	19	49·6	West of Udaigiri, Andhra Pradesh	8·68
4	June 20, 1968	09	28	32·1	Near Vijayawada, Andhra Pradesh	8·79

The apparent phase velocity of the first compressional wave from the main Bhadrachalam event had a weighted mean value of 8·96 km s^{-1}, which is quite consistent with the values obtained from all the thirteen aftershocks (Table 1). For a plane parallel layered crust, this would imply a Moho velocity (α_n) of 8·96 km s^{-1}. An independent analysis of four other events (Table 2) from distances between 225 km and 525 km in the azimuth range 30°–60° with respect to GBA gives a mean value of 8·82 km s^{-1} for α_n, in good agreement with the previous estimate. But both values are high compared with the global average of 8·11 km s^{-1} obtained from Herrin's revised travel times. But α_n obtained from near events recorded at GBA and which are at azimuths other than 30°–60° is 7·98 km s^{-1}, quite close to the global average. Further, one event west of Kadiri (June 9, 1968, onset at GBA 0023·03·2, distance 74 km and azimuth 49°) gave α_g (velocity of P_g phase) equal to 5·68 km s^{-1}, which is disproportionately small in relation to the anomalously high α_n obtained in this azimuth range. The available evidence shows therefore that the Moho-refracted phase, at least from the direction of Bhadrachalam, emerges at smaller angles compared with the angles of emergence in other azimuths. This could be due to non-uniform intermediate layers or to a Moho which dips by about 4° in the direction of Gauribidanur.

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Palaeomagnetism of some Middle Jurassic Lavas from South-east Argentina

In the classic papers on palaeomagnetism[1,2] the relative positions of South America and Africa in Gondwanaland are obtained from Palaeozoic data.

On the basis of palaeomagnetic studies[3–6] made with Mesozoic lavas of Argentina it has been suggested that in

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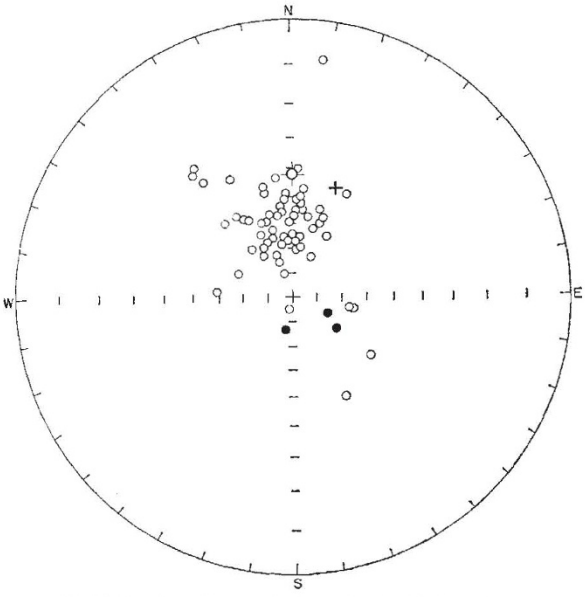


Fig. 1. NRM directions for samples from the middle Jurassic Chon Aike formation, Province of Santa Cruz, Argentina; downward inclinations are plotted as solid circles and upward inclinations as open circles.

Triassic times the relative positions of South America and Africa were similar to those obtained for the Palaeozoic but that the two continents had started to drift apart before the middle Cretaceous. It has been shown[7–9] that the Triassic poles of South America form a population which overlaps that of the African Mesozoic poles when these land masses are joined by their Atlantic littoral margins. The quantity and quality of our Mesozoic data have become a sound reason to justify the positions of South America and Africa in Gondwanaland.

Until now Jurassic palaeomagnetic poles for South America were not available. We made this study to answer the question of whether they belong to the Triassic population. An affirmative answer would suggest that the fragmentation of the South American–African block had not started in the Jurassic: a negative one would indicate that South America and Africa were drifting apart in the Jurassic.

To the best of our knowledge the result presented here is the first from a palaeomagnetic study made with Jurassic rocks of South America; an interpretation of this result is also given.

We have worked with keratophyres, quartz porphyries and porphyric rocks of middle Jurassic age of the Chon Aike formation. Dating by the potassium–argon method of a volcanic tuff confirmed the middle Jurassic age of this formation (160·7 m.y.)[10].

Hand samples from several stratigraphic levels were collected along the Atlantic coast, near Puerto Deseado (48° S, 66° W), Province of Santa Cruz.

Two specimens of each hand sample were subjected to a.c. cleaning while a third was used for thermal analysis. We found samples with normal and reversed natural magnetization, but the former predominated. Corrections to account for the original orientation of the bedding planes did not significantly affect the directions of the thermoremanent magnetization. The directions of the thermoremanent vector for each sample are plotted in Fig. 1. Table 1 shows the mean direction of the thermoremanent magnetization of the area together with the position of the middle Jurassic south pole calculated on the assumption of an axial dipolar field.

Fig. 2 shows the polar wandering curve for South America. The mean pole of the Triassic population (SA7, 82·6° S, 129° E, $\alpha = 9°$) was calculated on the basis of five of the available poles[5,7,8]: those which satisfy

Table 1. THERMOREMANENT MAGNETIZATION OF MIDDLE JURASSIC LAVAS (48° S, 68° W) (66 SAMPLES)

Declination	D	353°
Inclination	I	−63°
	R	61·37
Precision factor	K	14·03
Radius of confidence	α	4·8°
Pole position	84° S	56° E
Semi-minor axis	$d\psi$	6°
Semi-major axis	$d\chi$	7°

Irving's minimum criteria of reliability[11] with the additional constraint that the oval of confidence should not exceed 15°. Then the poles from Giron (Colombia) and Sierra de la Ventana (Argentina)[8] were not included. The circle of confidence of the middle Jurassic pole (SA8) overlaps a small part of the circle of confidence of SA7 and half of the area of the circle of confidence of the middle Cretaceous pole (SA9). Then, apparently, it would not be clear from the available data whether there is any real difference between the middle Jurassic pole and either the Triassic population or the middle Cretaceous pole. We think that this overlapping is usual when points on the polar wandering curve are determined from rocks not very different in age and the rate of polar wandering is not abnormally fast. The middle Jurassic pole is situated between the mean Triassic and the middle Cretaceous poles: SA7 (Triassic), SA8 (middle Jurassic 160·7 m.y.) and SA9 (middle Cretaceous, 115–125 m.y.) are three independent points in the South American polar wandering path. We cannot yet, however, discard the possibility that SA8 and SA9 represent the same population. We need more Jurassic–Cretaceous data to clarify this point, but if the possibility were confirmed, it would indicate a recurrent Mesozoic drift of South America and Africa[7], of the same characteristics as that quoted by Briden[2].

Fig. 3 shows a reconstruction of the relative positions of South America and Africa in Gondwanaland, based on

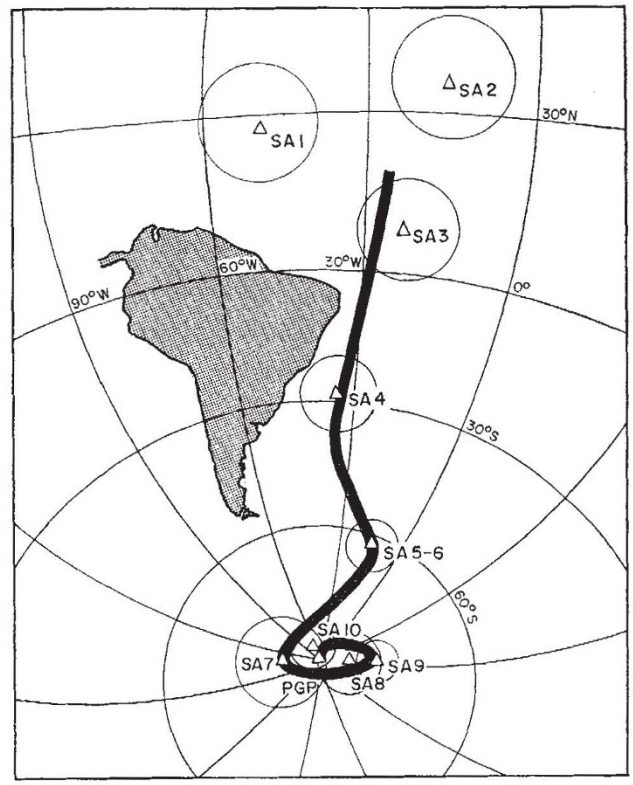


Fig. 2. Palaeomagnetic south poles and polar wandering path for South America: SA1 Cambrian; SA2 Silurian; SA3 Devonian; SA4 lower Carboniferous[5]; SA5–6 Permo-upper Carboniferous[9]; SA7 mean of the Triassic poles; SA8 middle Jurassic (160·7 m.y.); SA9 middle Cretaceous (115–125 m.y.)[1], and SA10 mean of the Cenozoic poles (0–26 m.y.)[12–13].

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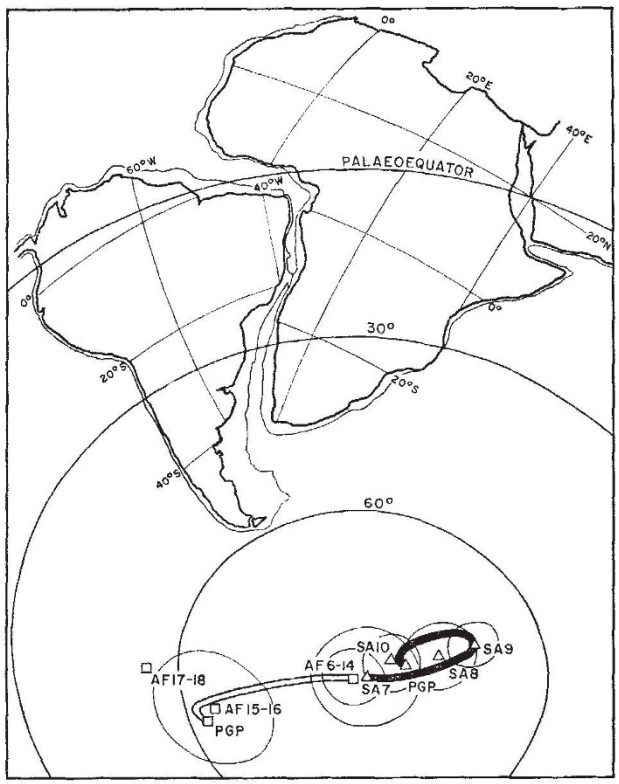


Fig. 3. Palaeomagnetic reconstruction of the positions of South America and Africa valid until the upper Triassic[7]. African poles are: Af 6–14 mean of Mesozoic poles; Af 15–16 mean of Tertiary poles and Af 17–18 mean of Quaternary poles[5]. PGP: present Geographic pole.

the Palaeozoic and Lower Mesozoic palaeomagnetic data[7]. For simplicity only the Mesozoic and Cenozoic palaeomagnetic data are shown. The circles of confidence of the mean of the African Mesozoic poles (Af 6–14) (ref. 5), and of the mean of the South American Triassic poles (SA7), overlap. The middle Jurassic pole of South America (SA8) is out of these populations, which indicates that in the middle Jurassic (160·7 m.y.) Africa and South America were drifting apart.

It can thus be inferred that the fragmentation of the South American–African block and the formation of the South Atlantic Ocean and the Argentine Basin began in Lower Jurassic times. This conclusion improves previous interpretations based on palaeomagnetic studies of South American rocks[3–9], and agrees pretty well with Palaeontological and geological evidence.

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Heat Flow Measurements on the Reykjanes Ridge

Heat flow measurements were made during a crossing of the North Atlantic from Nova Scotia to Iceland on R/V Trident of the University of Rhode Island, Cruise 41, in August–September 1967. Ten heat flow values were obtained on the crest of the Reykjanes Ridge near 60° N and 30° W. Temperature gradients were measured using the Ewing type thermograd instrument[1] modified at MIT by A. J. Erickson. The instrument consists of the temperature recorder and six thermistors: five of these are separated by 38 cm on the core barrel of a gravity corer. To measure the absolute water temperature, the sixth thermistor was mounted on the weight stand which remained above the water-sediment interface. The recorder was attached to the cable about 15 m above the weight stand. Three or four temperature probes, which penetrated 100–200 cm of the sediment, sufficed to determine the thermal gradient unambiguously (Fig. 1). The thermal conductivity of the sediments was measured by the needle probe method[2] on the ship after the sediments had reached room temperature and within 2 days of recovering the core. Mean thermal conductivity, obtained by averaging reciprocally the thermal conductivities measured every 10 cm on the relevant portion of the core, was combined with the thermal gradient to give the heat flow. The heat flow values were not corrected for the possible effect of seawater temperature fluctuation which was shown to be influential on heat flow in the Denmark Strait north-west of Iceland[3].

The results are summarized in Table 1 and Fig. 2. All the heat flow stations are located on the crestal zone of the Reykjanes Ridge, the northern extension of the Mid-Atlantic Ridge which lies south-west of Iceland. In this crestal zone, heat flow values vary considerably, the lowest being 0·80 μcalories/cm² s at station 8 and the highest 6·71 μcalories/cm² s at station 10. The second highest value, 5·68 μcalories/cm² s, occurs at station 15, some 26 miles west of station 10. The existence of the high heat flow zone over station 15 is confirmed by two adjacent stations, 14 and 16, which are more than 3 μcalories/cm² s. These high heat flows are separated from station 10 by two stations, 11 and 13, which have intermediate values, 1·47 and 1·74 μcalories/cm² s respectively, indicating that the two narrow high heat flow zones are symmetrically located on both sides of the crestal axis.

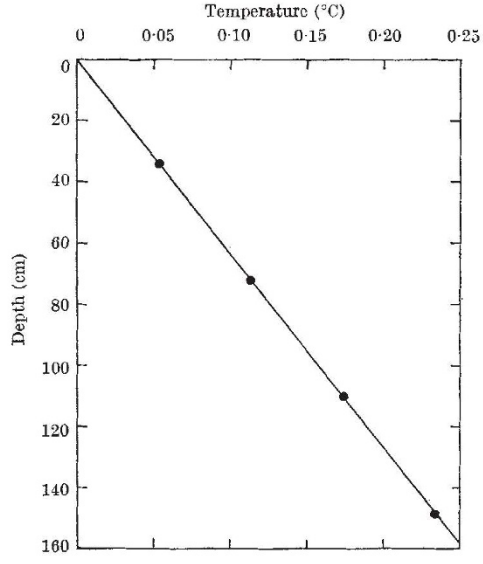


Fig. 1. Temperature versus depth for station TR41–14. The temperature plotted is the deviation from the bottom water temperature. Depth is measured from the top of the sediment.

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